Step 3: PPA structures and pricing maths (ticket PPA-3)

> **The parks on this page.** Two Encavis parks, modelled from public data: the Borrentin solar park (114 MWp, Mecklenburg-Western Pomerania, about 121 GWh a year) and the Everswinkel wind park (28.5 MW onshore, North Rhine-Westphalia, about 64 GWh a year). All numbers are from `reports/step03_ppa_structures.md`.
>
> **Time needed:** about 5 hours. Concepts take 90 minutes, code tour and findings 60 minutes, hands-on, notebook and exercises 2 hours, to-dos 30 minutes. **You will be able to:** write down the cash flows of every common PPA structure, explain which risk each one moves from producer to buyer, compute fair (break-even) prices, and back-test the structures on real data without fooling yourself about risk.

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0. The scenario

Wednesday. Origination comes back:

> *"The buyer for the Borrentin solar park wants options, not one price. Give me the economics of a pay-as-produced PPA, a baseload PPA, and a virtual PPA with a floor, plus the negative-price clause they keep asking about. And Risk wants a written methodology before anything goes to the credit committee. Ticket PPA-3."*

Two things are being asked. First, **structures**: who pays whom, when, and which risk each party carries. Second, a **methodology**: the maths written down so that Risk, Finance and the buyer can check it. Every number in the rest of this project is a cash flow of one of today's structures, evaluated on history today and on thousands of simulated futures from Step 6 on.

Ticket PPA-3: acceptance criteria (all met today)

- ☑ `pricing/contracts.py`: cash flows per interval for merchant, pay-as-produced (with and without a negative-price clause), virtual PPA (CfD), floor / cap / collar and baseload PPA; fair prices with discounting.
- ☑ Works on one history (pandas) and on many scenarios (NumPy arrays), for Step 7.
- ☑ Tests for the pricing invariants: a floor never lowers value, a CfD removes price risk, a baseload PPA is merchant plus a swap, the break-even strike is the capture price, and the fair price is a ratio of expectations.
- ☑ `METHODOLOGY.md` version 0: formulas, conventions, assumptions and limitations.

- ☑ Back-test 2024-2025 for the solar park (Borrentin) and the wind park (Everswinkel), strikes fixed from the previous year, with plan and surprise per structure (`ppa-lab ppa`).
- ☑ Learning page, notebook section, exercises.

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1. Concepts you need today

1.1 What a PPA is, in one paragraph

A power purchase agreement is a long-term contract (typically 5-15 years) between a producer and a buyer (a utility, a trader, or a corporate such as a chemicals or data-centre company) to sell electricity at an agreed price. The producer gets **bankable revenue** to finance the plant; the buyer gets **price certainty** and, with guarantees of origin, a credible green-power claim. A **physical** PPA delivers the energy (through balancing groups); a **virtual** (financial) PPA only settles a price difference while both sides keep trading on the spot market.

1.2 The cash flows, from the producer's side

Everything in `pricing/contracts.py` is one line of this table. P = spot price, q = energy produced in the interval, K = fixed price, B = band, F = floor, C = cap.

Structure	Cash flow per interval	Risk left with the producer
Merchant	$q \cdot P$	price level, shape, volume
Pay-as-produced (PaP) PPA	$K \cdot q$	volume
PaP, no payment at negative prices	$K \cdot q$, only when $P \geq 0$	volume, negative-price frequency
Virtual PPA (CfD)	$q \cdot P + (K - P) \cdot q = K \cdot q$	volume (and basis)
Floor	$q \cdot \max(P, F)$	upside price, shape; downside capped
Cap	$q \cdot \min(P, C)$	downside price; upside sold
Collar	$q \cdot \text{clip}(P, F, C)$	price only inside [F, C]
Baseload PPA	$K_B \cdot B + (q - B) \cdot P$	**shape** and volume

Work through the running example from `tests/test_contracts.py` by hand. Four intervals, a solar-like profile that produces in the cheap ones:

	t1	t2	t3	t4	Total
Price P (EUR/MWh)	50	−10	100	20	base 40
Energy q (MWh)	2	4	0	4	10
Merchant q·P	100	−40	0	80	**140** (capture 14)
PaP at 30	60	120	0	120	300
PaP at 30, no pay below 0	60	0	0	120	180
Floor 0: q·max(P, 0)	100	0	0	80	180
Baseload at 40, B = 2.5	75	85	−150	130	**140**

Look at t3: the baseload seller must deliver 2.5 MWh it does not have, buys it at 100 EUR/MWh and pays 150 EUR net. That single cell is the essence of shape risk.

1.3 Pay-as-produced: the buyer takes price and shape

Under a PaP PPA the buyer takes every MWh the plant produces, whenever it produces, at K. The producer's revenue is K · (total energy): **no price risk and no shape risk left**, only volume risk (a poor solar year means less energy at the same K).

The buyer's view is the mirror image: it pays K for energy worth P at that moment, so its gain per MWh against buying on the spot market is (P – K). Buying solar-shaped energy at a fixed price means the buyer owns the cannibalisation problem from Step 2. That is why the fair PaP price is the **capture price**, not the base price.

1.4 Baseload PPA: fixed level, but the shape stays with you

A baseload PPA promises a constant band B every hour. The solar park produces more than B at midday in summer (it sells the surplus at low prices) and less than B at night and in winter (it buys the shortfall, often at high prices). Rearranging the cash flow:

```
K_B · B + (q – B) · P   =   q · P   +   B · (K_B – P)
   baseload PPA             merchant     baseload swap
```

A baseload PPA is merchant revenue plus a fixed-for-floating swap on a flat band. It fixes the price *level* of the band, but the producer still earns its own capture price on its own profile. If base prices rise while solar's capture rate falls, the swap loses and the merchant leg does not make it up. That is exactly what happened to the solar park in 2025 (section 4.2).

> **Interview:** "Why is a baseload PPA priced differently?" Because the producer must buy its shortfalls at whatever the market charges, and for solar those hours are the expensive ones: volume and price are negatively correlated. The residual position is short the shape, and the producer prices that risk in.

1.5 Virtual PPA: a CfD that settles the difference

In a virtual PPA nothing is delivered. The producer sells on the spot market as usual, and the two parties settle (K – P) · q: the buyer pays when the price is below K and receives when it is above. Net for the producer: K · q, the same as a physical PaP.

What differs is not the economics but the plumbing:

- **Basis risk:** the settlement uses a reference price (here the DE-LU day-ahead price) that may differ from the price the plant actually achieves (intraday, imbalance, other bidding zones in multi-country deals).
- **Accounting:** a contract that settles net in cash can count as a derivative and may need fair-value accounting; physical PPAs often fall under an own-use exemption. Finance will ask about it.
- **Volume definition:** "as produced" (our case), "baseload volume", or a shaped profile.

1.6 Floors, caps and collars are options

A floor F pays max(F – P, 0) per MWh: the producer owns a strip of **put options**, one per interval. A cap C costs the producer max(P – C, 0) per MWh: it **sells calls**. A collar combines both; a **zero-cost collar** chooses F and C so that the premium paid for the floor equals the premium received for the cap.

Two things to remember:

- **A floor at 0 EUR/MWh is the right to curtail.** A plant that can switch off when the price is negative earns q · max(P, 0), exactly the cash flow of owning a 0-strike floor. Flexibility is an option, and options have value.
- **Option payoffs are convex.** Raising a floor by 10 EUR/MWh adds more and more value as the floor moves into the bulk of the price distribution (exercise 2 shows the curve). This is why Step 7 needs Monte Carlo: the value depends on the whole distribution, not on the average price.

Today's cash flows show the **payoffs** only. What the option **costs** (its premium) is its expected discounted payoff over simulated scenarios, which comes in Step 7.

1.7 Negative-price clauses

With 24% of solar energy produced at negative prices in 2025 (Step 2), every term sheet says what happens then. The common version: the buyer **does not pay** for energy produced while the price is below zero, and the producer curtails. For the producer that removes a slice of paid volume, so the strike must go up to stay whole. How much is exercise 3; spoiler: much more for solar than for wind, for the reason you saw in Step 2. A price of exactly zero is still paid: EEG §51 and typical clauses refer to prices *below* zero (`test_a_price_of_exactly_zero_is_not_negative`).

1.8 Fair prices: the maths in three lines

The fair (break-even) strike makes the contract worth the same as selling at spot, in expectation over scenarios (price paths and weather years), with discount factors D:

```
K*    = E[ sum_t D_t · q_t · P_t ] / E[ sum_t D_t · q_t ]       pay-as-produced
K_B*  = E[ sum_t D_t · P_t ]       / sum_t D_t                  baseload
D_t   = (1 + r) ^ (-years from the valuation date)
```

Two traps, both tested:

- K *is a ratio of expectations**, not the average of each scenario's capture price: a windy year with lots of energy weighs more than a calm one (`test_fair_price_is_a_ratio_of_expectations_not_an_average_of_ratios`).
- With one history and no discounting, K *is the capture price and K_B* the base price, and at those strikes every structure earns exactly the merchant revenue in total (`test_at_the_fair_price_every_structure_earns_the_merchant_revenue_in_total`).

Why the capture price sits below the base price, in one identity (METHODOLOGY.md §2):

```
capture price = base price + Cov(q, P) / mean(q)
```

Solar output and price move in opposite directions, the covariance is negative, and the capture price falls below base. Step 2's cannibalisation chart is this covariance.

From a market quote to an offer, the stack your preparation guide sketches (Figure 5):

```
PaP price ≈ baseload forward × expected capture rate
            − profile & volume risk premium − negative prices & curtailment
            − balancing & route to market − credit & liquidity
e.g.  85 × 0.55 ≈ 47  − 3 − 1.5 − 1.5 − 1  ≈ 40 EUR/MWh   (illustrative only)
```

Today implements the first term on history; Steps 5-7 replace "expected capture rate" by a simulated distribution and compute the premia.

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2. What was built: architecture

```
data/processed/prices_native.parquet  ─┐
data/processed/power_15min.parquet    ─┤
                                       ▼
analysis/capture.py: prices_on_grid()   (Step 2: prices on the 15-min grid)
                                       │
pricing/backtest.py                    ▼
  park_energy()   national profile → Borrentin 114 MWp × 1,061 h, Everswinkel
                  28.5 MW × 2,246 h (their published yields), scaled per year
  site_energy()   or the park's own Step 4 profile (report section 2)
  backtest()      strikes from Y-1, cash flows in Y, plan = same terms on Y-1
  summarise()     revenue, per MWh, vs merchant, plan, surprise, worst month
        │ uses
        ▼
pricing/contracts.py   merchant · pay_as_produced · cfd · collar · baseload
                       fair_pay_as_produced_price · fair_baseload_price · discount_factors
        │
        ▼
analysis/step03_report.py → reports/step03_ppa_structures.md + 2 charts   (`ppa-lab ppa`)
METHODOLOGY.md            the maths, conventions, assumptions and limitations (v0)
config/project.toml       [pricing]: parks, delivery years, floor
```

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3. Code tour

3.1 `pricing/contracts.py`: one line per structure, on purpose

- **Cash flows per interval, not totals.** Each function returns an array (or Series) shaped like its inputs. The caller decides how to aggregate: by month for a report, by year for a back-test, by scenario for Monte Carlo. One function, three uses.
- **pandas or NumPy.** A Series in gives a Series out with the same index (so you can `groupby` months); arrays of shape (scenarios, intervals) work unchanged for Step 7.

- **Refuses mismatched time grids.** pandas aligns two Series on their index and fills gaps with NaN without a word. A one-hour shift (a time-zone bug) would silently produce NaN sums. `_check_same_grid` raises instead (`test_contracts_refuse_series_on_different_time_grids`).
- **Producer perspective everywhere.** One sign convention, stated in the module docstring and in METHODOLOGY.md. Mixing buyer and seller signs is a classic source of "the PPA is worth minus three million" surprises.
- **No premiums inside cash flows.** A floor's cash flow is its payoff; what it costs is a separate valuation question.

3.2 `pricing/backtest.py`: an honest back-test

- `park_energy` scales the national profile per year to capacity × full-load hours. Scaling per year removes the national fleet's growth (solar grew 30% from 2023 to 2025), which a single park would not see, but keeps the real hour-by-hour shape.
- **No hindsight.** Strikes for year Y come from year Y−1 only (`test_strikes_are_fixed_at_the_previous_years_outcome`; a mutation that used the delivery year itself was caught).
- **Plan and surprise.** The plan applies the same contract terms to Y−1's prices and profile: what the producer expected when it signed. Realised minus plan is the risk the structure left with it. Section 4.1 explains why this replaced a first idea.

3.3 Configuration and command line

`config/project.toml` has a new `[pricing]` section: the two parks, the delivery years and the floor. The config loader checks that each delivery year **and the year before** are complete years inside the data range, and fails early otherwise. `ppa-lab ppa` writes the report; inside `ppa-lab all` it **skips with a warning** when the data range is too short (the two-day test fixtures), instead of breaking the whole pipeline.

3.4 Tests: what each protects

Test file	Protects against
`test_contracts.py` (18)	floor coded as a cap; baseload residual with the wrong sign; average of ratios instead of ratio of expectations; zero price treated as negative; silent NaNs from misaligned indexes
`test_backtest.py` (8)	hindsight strikes; one scale factor for all years; wrong plan; wrong units
`test_step03_report.py` (1)	the full report on two synthetic years (all charts and tables)
`test_config.py` (+2), `test_cli.py` (+1)	invalid delivery years; `ppa` command and graceful skip in `all`

Each key test was checked by mutation: the planted bug had to make it fail.

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4. Four real findings

4.1 My first risk metric was wrong, and the data said so

- **First version:** the report showed the standard deviation of monthly revenue within each year as "risk".
- **Symptom:** for the solar park the pay-as-produced PPA came out **riskier** than merchant (monthly std 269k vs 171k EUR in 2025). A fixed-price contract cannot carry more price risk than the spot market. Something was measuring the wrong thing.
- **Diagnosis:** the spread across months within a year is mostly **seasonality**. Under a PaP PPA, revenue follows volume: big in summer, small in winter. Merchant revenue is smoother because the high summer volume meets low summer prices (a natural hedge). Seasonality is known in advance: it is a liquidity topic, not risk.
- **Fix:** risk is the **surprise against what you expected** when signing. The plan applies the structure to the previous year at the same terms; the surprise is realised – plan. Result: the PaP PPA has a surprise of exactly 0.00 EUR/MWh in every case, as it should. The monthly chart stays in the report, relabelled "seasonality and liquidity".
- **Lesson:** when a number contradicts what the contract obviously does, question the metric before the contract. In an interview this is worth more than any formula.

4.2 Same baseload PPA, opposite outcomes

Both parks signed a 2025 baseload PPA at 78.51 EUR/MWh (the 2024 base price). The 2025 base came in at 89.32, so the swap leg lost 10.81 EUR/MWh for both. But:

2025	Merchant surprise	Swap leg	Baseload surprise
Solar park	−0.53	−10.81	**−11.34**
Wind park	+12.73	−10.81	**+1.92**

Why is explained in exercise 4: write it yourself before you look at the notebook.

4.3 A baseload seller can have negative months

The solar park's baseload PPA had a **negative month in January 2025** (−58k EUR): the 13.8 MW band had to be delivered while the park produced almost nothing, and the shortfall was bought at winter prices. For a project company with a debt service schedule, a negative cash month is a real problem even if the year is fine. That is why lenders look at monthly cash flows and why "worst month" is in the report.

4.4 Small ones

- NumPy 2 removed the `ndarray.ptp()` method; `np.ptp(array)` is the replacement.
- The two-day CLI test exposed a design question: should `ppa-lab all` fail when the data cannot support a back-test? No: it skips Step 3 with a clear warning, and the config validator catches impossible delivery years in real configurations.

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5. Hands-on: run it yourself

```
cd ".../Encavis quantitative/ppa-lab"
make test                  # the offline tests (290 in v1.11)
make ppa                   # reports/step03_ppa_structures.md + 2 charts
open METHODOLOGY.md        # the maths, rendered on GitHub
make notebook              # Step 3 section: 3.1 to 3.6, then your solutions
```

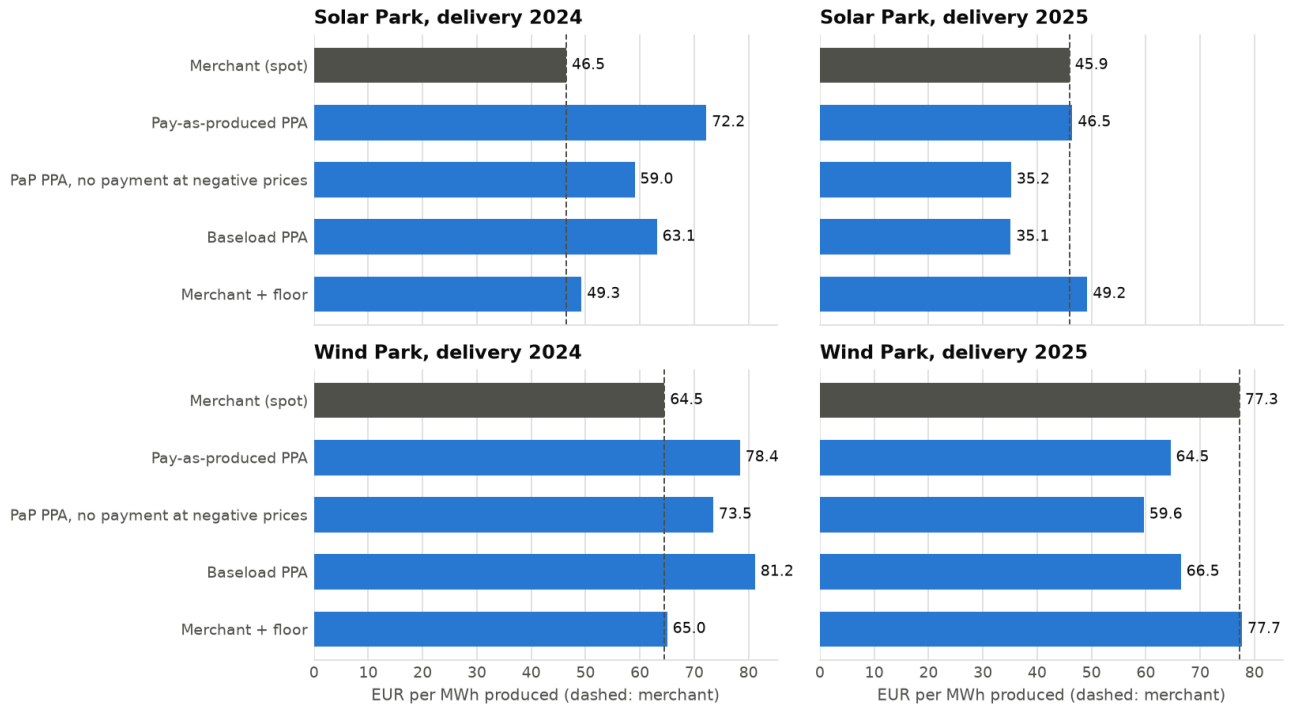
In the notebook, start with 3.1 (the running example as a table), then play with 3.2 (one day, every structure, quarter by quarter) and 3.5 (your own contract terms).

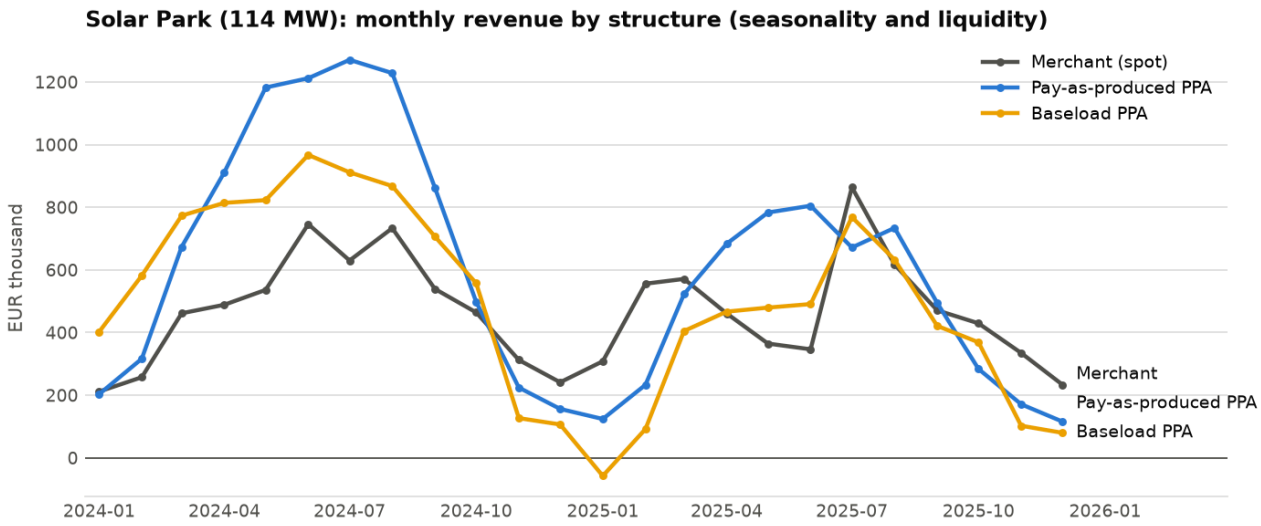
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6. Results: what the back-test says

6.1 Revenue per MWh produced, by structure (EUR/MWh)

Park, year	Merchant	PaP	PaP, no pay < 0	Baseload	Merchant + floor 0
Solar 2024 (strikes 72.25 / 95.18)	46.48	**72.25**	59.04	63.14	49.26
Solar 2025 (strikes 46.48 / 78.51)	45.95	46.48	35.22	**35.14**	49.25
Wind 2024 (strikes 78.44 / 95.18)	64.54	78.44	73.50	**81.20**	65.03
Wind 2025 (strikes 64.54 / 78.51)	**77.27**	64.54	59.63	66.46	77.71





The table uses the national profile of each technology, so its EUR/MWh do not depend on the park's size. With each park's own modelled output (ERA5 weather at the site, Step 4; report section 2) Borrentin earned 47.22 merchant and 76.06 under the PaP PPA in 2024 (national profile: 46.48 and 72.25), and Everswinkel 76.34 merchant in 2025 (77.27): the sites earn close to the national profiles, Borrentin slightly more, Everswinkel slightly less.

6.2 Five stories

1. **Fixing a price is a bet on direction, even when it is "fair".** Signed at the 2023 capture price, the solar PaP PPA earned 72.25 in 2024 against 46.48 merchant: +EUR 3.1 million for Borrentin's 121 GWh. The wind PaP signed at the 2024 capture price gave up 12.73 EUR/MWh in 2025 when prices recovered. The PPA did its job both times: it removed the surprise. Whether it "won" is hindsight.
2. **The negative-price clause is expensive for solar.** In 2025 it cut the solar PaP from 46.48 to 35.22 EUR/MWh, a loss of 11.26 EUR/MWh: 24% of the energy was not paid. For wind it cost 4.91. A clause like this must be priced into the strike.
3. **Curtailment is worth real money.** A floor at 0 (or simply switching off at negative prices) added 3.30 EUR/MWh for solar in 2025 and 0.44 for wind.
4. **Baseload hedges the level, not the shape.** Solar 2025: −11.34 EUR/MWh surprise under the baseload PPA, worse than merchant (−0.53). The level rose, solar's share of it did not (section 4.2).
5. **Liquidity is not the same as value.** The solar baseload PPA had a negative month (January 2025); the PaP PPA had its weakest months in winter; merchant in spring. Lenders care about all three.

6.3 What this means for pricing the solar PPA

The fair PaP price is anchored on the capture price (about 46 EUR/MWh on 2024-2025 data), not on the base price (78-89). A negative-price clause raises the break-even strike substantially, and a baseload structure leaves the producer short the shape: if the buyer wants a baseload product from a solar park, the producer must charge for that risk or firm the profile (storage, a wind portfolio, or buying back shape on the market). Step 7 puts numbers on these risk premia.

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7. Exercises

Warm-up (in `exercises/step03_ppa.py`, checked by `.venv/bin/pytest exercises -q`)

1. **A baseload PPA by hand.** Revenue per MWh for solar and wind in 2025 at 78.51.
2. **What is a floor worth?** For floors at 0 and 20 EUR/MWh in 2025.
3. **Break-even strike with a negative-price clause.** By how much does the clause raise the fair strike for solar and for wind?
4. **Written:** same baseload contract, opposite outcomes (section 4.2). Why?

Core

5. **Monthly floor (test first).** Many floors settle on the **monthly average** capture price, not per interval. Write `monthly_floor_value(prices, volumes, floor, tz)` in a new test-first function and compare it with the interval floor for solar 2025. Which is worth more, and why (think about averaging and convexity)?
6. **Zero-cost collar.** For solar 2025, find the cap C that makes a collar with floor 20 cost nothing on history (floor payout = cap payout). Use a simple bisection.
7. **Hybrid park.** Combine the solar and wind parks into one portfolio and compute the portfolio's capture rate and the baseload PPA surprise for 2025. Does the portfolio suit a baseload product better than solar alone?

Stretch

8. **Discounting over ten years.** Build a 10-year PaP with flat expected prices and a 6% discount rate. Show that discounting changes the fair price only if volumes or prices are not flat over time (hint: the ratio of expectations).
9. **Buyer's view.** Write the buyer's P&L for each structure and check that producer P&L + buyer P&L = 0 against the spot alternative.

▼ **Hints and solution sketches**

1. B = q.sum() / len(q); cash = K·B + (q − B)·P; divide the sum by q.sum(). Solar is far below merchant, wind slightly below.
2. (floor − p).clip(lower=0) times q, summed, divided by q.sum(). Raise the floor and watch the value grow faster than linearly.
3. Solve K · (energy at P ≥ 0) = merchant revenue. Compare with Step 2's capture price.
4. Split each baseload surprise into "merchant surprise" + "swap leg"; the swap leg is the same for both parks; the merchant surprise is not. What drives each park's merchant surprise: the level, the shape, or both?
5. Group by month: payout_m = q_m · max(F − capture_m, 0). By Jensen's inequality the average of payoffs is at least the payoff of the average: the interval floor is worth more.
6. Write f(C) = floor payout − cap payout; f decreases in C; bisect between 20 and 500.
7. Sum the two energy series; the portfolio's profile is flatter, so the residual is smaller.
8. With flat q and P, the weights D_t·q_t are proportional for numerator and denominator.
9. Buyer of a physical PaP: (P − K) · q per interval.

8. Self-check quiz

1. Write the producer's cash flow of a PaP PPA, a CfD and a baseload PPA.
2. Why do a physical PaP PPA and a virtual PPA give the producer the same cash flow?
3. Which risk does a PaP PPA leave with the producer, and which does a baseload PPA leave?
4. Show that a baseload PPA equals merchant revenue plus a swap.
5. Why is a floor at 0 EUR/MWh the same as the right to curtail?
6. Why is the fair PaP price a ratio of expectations and not an average of capture prices?
7. What does a negative-price clause do to the fair strike, and why more for solar?
8. Why was the monthly revenue spread a bad risk measure, and what replaced it?
9. What is the "plan" in the back-test, and why must strikes come from the year before?
10. What is missing between the historical fair price and a price you would quote?

▼ **Answers**

1. PaP: $K \cdot q$. CfD: $q \cdot P + (K - P) \cdot q = K \cdot q$. Baseload: $K_B \cdot B + (q - B) \cdot P$.
2. In the virtual PPA the producer sells at spot and receives $(K - P) \cdot q$ from the buyer; the price terms cancel. Differences are basis, volume definition and accounting.
3. PaP: volume only. Baseload: shape and volume (the price level of the band is fixed).
4. $K_B \cdot B + (q - B) \cdot P = q \cdot P + B \cdot (K_B - P)$: merchant plus a fixed-for-floating swap on B.
5. Curtailing at $P < 0$ earns $q \cdot \max(P, 0)$, the payoff of spot plus a 0-strike put.
6. Scenarios with more energy contribute more revenue and more MWh; averaging capture prices would give a calm year the same weight as a windy one.
7. It removes paid volume in negative hours, so K must rise to keep revenue equal to merchant; solar produces far more of its energy at negative prices (24% in 2025).
8. Within a year it mostly measures seasonality, which is known in advance; the surprise against plan (realised – expected at signing) measures risk.
9. The same contract terms applied to the previous year's prices and profile: the expectation at signing. Using delivery-year data for the strike would be hindsight.
10. Forward prices instead of last year's outcome, a distribution instead of one history, risk premia, negative-price and curtailment costs, balancing, route to market, credit and discounting over the full tenor.

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9. Interview corner

Your 60-second story (STAR):

> *Situation:* a buyer asked for alternatives to a pay-as-produced solar PPA. *Task:* show the economics and the risk transfer of each structure and write the methodology for Risk. *Action:* I implemented the cash flows of merchant, pay-as-produced, virtual PPA, floors and collars and baseload PPAs as vectorised functions, tested their invariants (a floor never lowers value, a baseload PPA is merchant plus a swap) and back-tested them on 2024-2025 with strikes fixed from the previous year. *Result:* the PaP PPA removed all price surprise; the baseload PPA on solar lost 11 EUR/MWh against plan in 2025 because it hedged the level but not the shape; and a negative-price clause cost 11 EUR/MWh.

> I also caught my own wrong risk metric: monthly volatility was seasonality, so I replaced it with surprise against plan.

Likely questions and strong answers:

- "*How would you price a 10-year PaP PPA for a German solar park?*" Baseload forward curve times a simulated capture rate (joint weather and price scenarios), fair price as E[sum D·q·P] / E[sum D·q], then risk premia from the distribution (P5 or CFaR), minus negative-price, balancing, route-to-market and credit costs; sanity-check against market values.
- "*Why is a baseload PPA priced differently?*" The producer delivers a flat band and buys its shortfalls when it does not produce; for solar those are the expensive hours, so it is short the shape. Baseload PPA = merchant + swap.
- "*What is a floor worth?*" The expected discounted payoff of a strip of puts; convex in the strike, so it needs the price distribution, not the average. A 0-strike floor is the value of curtailment.
- "*The buyer wants no payment at negative prices. What do you do?*" Quantify the share of energy at negative prices in the scenarios, raise the strike to keep value (substantially for solar), or offer curtailment rights or a battery to shift energy.

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10. How this maps to real work at Encavis

In this project	In a company
`contracts.py` cash flows	The pricing library behind every term sheet and the deal model
Prior-year strikes	Forward curves from EEX, broker quotes, the hourly price forward curve
One history	Monte Carlo over price paths and weather years (Steps 5-7)
Plan and surprise	Budget vs actual reporting, PPA performance monitoring
METHODOLOGY.md	Model documentation for Risk, auditors and model validation
Negative-price clause, floors	Term-sheet negotiation with offtakers; structuring desk
Worst month	Debt sizing and liquidity planning with lenders

Encavis sells part of its output through PPAs (and runs merchant and regulated assets too); each new PPA starts with exactly these structure comparisons.

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11. Your to-dos for Step 3

- □ Read sections 1 and 4 with a notebook; redo the running example (1.2) by hand.
- □ Read `METHODOLOGY.md` once on GitHub (formulas render there).
- □ Run section 5; play with notebook cells 3.2 and 3.5.
- □ Exercises 1-4 in `exercises/step03_ppa.py`; check them in the notebook's "My solutions: Step 3" block.
- □ **Commit Step 3** (commands below).

- □ **German vocabulary of this step:** der Stromabnahmevertrag (PPA), erzeugungsabhängig (pay-as-produced), die Bandlieferung (baseload), der Differenzvertrag (CfD), die Preisuntergrenze (floor), die Preisobergrenze (cap), der Ausübungspreis (strike), die Gegenpartei, das Ausfallrisiko (credit risk), die Liquidität.
- □ Stand-up update: "*Yesterday: PPA structures and methodology v0; back-test shows a baseload PPA on solar leaves shape risk (–11 EUR/MWh in 2025). Today: weather to power (PPA-4). Blockers: none.*"

Commit (check `git status` first: no `.env`, no `M .env.example`):

```
cd ".../Encavis quantitative/ppa-lab"
git status --short
git add src tests exercises config reports docs notebooks METHODOLOGY.md README.md
CHANGELOG.md Makefile pyproject.toml
git status --short
git commit -m "feat(pricing): PPA structures, fair prices and 2024–2025 back-test (PPA-3)"
git push
gh run watch
```

---

12. Glossary

Term	Meaning
PPA	Power purchase agreement: long-term contract to buy or sell electricity
Pay-as-produced	The buyer takes the actual output at a fixed price
Baseload PPA	Constant band every hour; producer settles the residual at spot
Virtual PPA / CfD	Financial contract settling (K – P) on a volume; no delivery
Strike	The fixed price K of a PPA, CfD or option
Floor / cap / collar	Minimum / maximum price, or both; strips of puts / calls
Zero-cost collar	Floor and cap chosen so their premiums offset
Shape risk	Risk from when energy is produced vs when it is expensive
Basis risk	Difference between the settlement price and the achieved price
Fair (break-even) price	Strike making the contract worth the same as merchant
Plan / surprise	Expected outcome at signing / realised minus plan
CFaR	Cash flow at risk: expected cash flow minus a low quantile

13. Sources

- Encavis preparation guide (7 Oct 2026): pricing equations, Figure 5 pricing stack, interview questions, project specification (sections 3.3, 5, 6.2)
- AFRY, "Correctly valuing baseload PPAs in the Nordic market" (guide source S14)
- EEG 2023 §51 (no payment entitlement in periods with negative prices)

- Step 2 results: capture prices and negative-price shares 2023-2025 (`reports/step02_capture_prices.md`)